

Early Identification of Heart Attack vs Gastritis Using AI/ML

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Project Proposal Report

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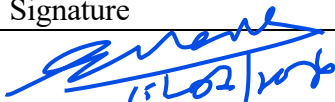
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Chest pain is one of the most frequently reported symptoms in emergency departments and clinical settings, and accurately identifying its underlying cause is critical for effective treatment. Cardiac-related chest pain caused by conditions such as myocardial infarction or coronary artery disease requires immediate medical attention. However, early identification of cardiac abnormalities remains challenging because symptoms may overlap with non-cardiac conditions. Electrocardiography (ECG) and echocardiography are two of the most widely used diagnostic tools for evaluating cardiac conditions, yet the manual interpretation of these diagnostic tests requires significant clinical expertise and may be time-consuming in high-demand healthcare environments. Although recent advances in artificial intelligence have enabled the development of automated systems for analyzing medical signals and images, many existing approaches focus primarily on single diagnostic modalities such as ECG image analysis or echocardiographic image interpretation. These isolated approaches may limit diagnostic reliability because they do not provide cross-validation between different cardiac diagnostic techniques. Therefore, there is a need for an intelligent system that integrates multiple cardiac diagnostic inputs to improve early detection and diagnostic confidence. This research proposes an AI-assisted cardiac diagnostic component that analyzes ECG images to identify abnormal cardiac patterns and subsequently validates the findings through echocardiographic image analysis. The ECG analysis model uses deep learning techniques to detect waveform abnormalities associated with cardiac conditions, while the echocardiography model evaluates structural and functional heart characteristics to confirm potential abnormalities. The system employs convolutional neural network (CNN) architectures for medical image classification, with preprocessing techniques applied to enhance ECG waveform visibility and echocardiographic image quality. Tools such as Python, PyTorch, and OpenCV are utilized for model development, training, and image processing. The expected outcome of this research is the development of a reliable AI-assisted cardiac analysis component that supports clinical decision-making by providing automated interpretation of ECG and echocardiographic data. The system aims to improve early detection of cardiac abnormalities and reduce the burden of manual diagnostic interpretation. From a social perspective, the proposed system can contribute to improved patient outcomes by enabling faster identification of heart-related conditions. From a commercial perspective, technology has potential applications in healthcare diagnostic platforms, telemedicine systems, and AI-based medical decision-support tools.

Keywords: Artificial Intelligence, ECG Image Analysis, Echocardiography, Cardiac Abnormality Detection, Clinical Decision Support.

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LIST OF ABBREVIATIONS

- 2D-CNN: 2-Dimensional Convolutional Neural Network
- 3D-CNN: 3-Dimensional Convolutional Neural Network
- ACS: Acute Coronary Syndrome
- AI: Artificial Intelligence
- API: Application Programming Interface
- AUC: Area Under the Curve
- CNN: Convolutional Neural Network
- CVD: Cardiovascular Disease
- ECG: Electrocardiogram
- Echo: Echocardiogram
- ER: Emergency Room
- F1-Score: Harmonic mean of precision and recall
- GERD: Gastroesophageal Reflux Disease
- GPU: Graphics Processing Unit
- IEEE: Institute of Electrical and Electronics Engineers
- LKR: Sri Lankan Rupee
- AUC-ROC: Area Under the Receiver Operating Characteristic
- SDG: Sustainable Development Goal
- ROC: Receiver Operating Characteristic
- SaaS: Software as a Service
- WHO: World Health Organization
- WMA: Wall Motion Abnormality
- STEMI: ST-Elevation Myocardial Infarction
- NSTEMI: Non-ST-Elevation Myocardial Infarction
- UI: User Interface
- OpenCV: Open-Source Computer Vision Library

LIST OF APPENDICES

Appendix	Title
Appendix A	Survey Instruments
Appendix B	Preliminary UI Mockups
Appendix C	Risk Analysis
Appendix D	Ethical Documents

Table 1: List of Appendices

1. INTRODUCTION

1.1 Background and Context

Cardiovascular diseases remain one of the leading causes of mortality worldwide, with acute myocardial infarction and other cardiac disorders accounting for a significant proportion of global deaths. Early diagnosis of cardiac conditions is essential because delayed detection can lead to severe complications or fatal outcomes. However, a major clinical challenge arises when cardiac symptoms mimic gastrointestinal disorders such as gastritis, gastroesophageal reflux disease (GERD), or gastric pain. Patients often present with symptoms such as epigastric pain, nausea, vomiting, or burning sensations in the chest, which are commonly associated with gastric conditions but may also indicate an underlying heart attack. This overlap in symptom presentation frequently leads to misinterpretation and delayed diagnosis of life-threatening cardiac conditions [1].

Several clinical cases and medical reports have documented situations where patients presenting with gastric-like symptoms were initially diagnosed with gastrointestinal diseases but later discovered to have myocardial infarction after undergoing electrocardiogram (ECG) testing. In some cases, failure to conduct early ECG evaluation resulted in delayed treatment and even fatal outcomes. Medical investigations emphasize that atypical myocardial infarction can manifest without classic chest pain and may instead present as abdominal or gastric discomfort, highlighting the importance of comprehensive cardiac evaluation in patients with unexplained epigastric symptoms [2]. Studies analyzing chest pain cases in primary care settings have also demonstrated that gastrointestinal conditions can mimic cardiac symptoms, making it difficult for clinicians to differentiate between cardiac and non-cardiac causes without proper diagnostic testing [3].

Electrocardiography (ECG) and echocardiography (Echo) are two critical diagnostic tools widely used in cardiovascular assessment. ECG measures the electrical activity of the heart and helps detect abnormalities such as arrhythmias, ischemia, and ST-segment changes associated with myocardial infarction. Echocardiography provides ultrasound imaging of the heart and enables clinicians to observe structural and functional abnormalities such as ventricular dysfunction or wall motion defects. These diagnostic methods are essential for confirming cardiac abnormalities when symptoms alone are insufficient for accurate diagnosis [4]. However, manual interpretation of ECG image and echocardiographic images requires substantial clinical expertise and may be prone to human error or variability in interpretation.

In many healthcare environments, particularly in developing countries, ECG results are commonly provided as printed charts or scanned digital images rather than raw digital signals. Therefore, automated analysis of ECG report images has become an important research area in medical artificial intelligence. Image-based ECG analysis allows deep learning models to interpret waveform patterns directly from clinical ECG reports, enabling practical deployment in healthcare systems where signal-level data is not readily available.

Recent developments in artificial intelligence and deep learning have significantly improved automated medical diagnostic systems. Deep learning models, particularly 2D-CNNs, have

demonstrated strong performance in analyzing medical images and biomedical data. In ECG-based diagnosis, 2D-CNN models can analyze ECG waveform report images and automatically learn spatial patterns such as ST-segment elevation, waveform morphology, and rhythm abnormalities. Studies have shown that AI-based ECG analysis systems can identify complex cardiac patterns with high accuracy by extracting hierarchical features directly from ECG images [5].

Similarly, echocardiography consists of ultrasound video sequences that capture the motion of cardiac structures over time. Therefore, 3D-CNNs are particularly suitable for echocardiographic analysis because they can capture both spatial and temporal features from sequential image frames. These models enable automated detection of structural abnormalities, ventricular dysfunction, and abnormal heart motion patterns in echocardiographic data [6].

Within the proposed research system, Component 2 focuses on the automated analysis of ECG and echocardiographic data as part of a multi-stage diagnostic workflow. During the secondary validation stage, ECG report images are analyzed using a 2D-CNN to detect electrical abnormalities associated with cardiac conditions. Patients showing abnormal ECG findings are then referred for echocardiographic analysis using a 3D-CNN, which analyzes ultrasound image sequences to identify structural heart abnormalities and abnormal cardiac motion patterns. By combining these two complementary diagnostic modalities, the proposed system aims to improve diagnostic reliability and reduce the risk of misdiagnosis.

This research aligns with the United Nations Sustainable Development Goal 3 (SDG 3): Good Health and Well-Being, which emphasizes reducing premature mortality from non-communicable diseases such as cardiovascular diseases. By leveraging artificial intelligence to improve early cardiac detection and diagnostic accuracy, the proposed system contributes to strengthening digital healthcare technologies and enhancing clinical decision support systems. Furthermore, the study aligns with the Artificial Intelligence and Digital Health research cluster, which focuses on applying machine learning and intelligent systems to address critical healthcare challenges and improve patient outcomes.

1.2 Literature Review

Recent research has increasingly focused on applying artificial intelligence and machine learning techniques to improve cardiovascular disease detection. Electrocardiography has been widely studied as a key diagnostic tool for detecting heart abnormalities because it records the electrical activity of the heart and provides important information about cardiac rhythm and conduction patterns. Traditional ECG interpretation methods rely heavily on manual analysis by clinicians, which may lead to diagnostic variability and time-consuming processes. To overcome these limitations, many studies have explored automated ECG classification systems using deep learning models such as convolutional neural networks (CNNs) [7].

Deep learning-based ECG analysis has demonstrated strong potential for detecting cardiac abnormalities with high accuracy. For example, studies using convolutional neural networks (CNNs) have achieved high classification performance in detecting cardiac abnormalities using large ECG datasets such as the PTB-XL ECG Dataset and other publicly available ECG

waveform image datasets. These models automatically learn hierarchical features from ECG images and eliminate the need for manual feature extraction, making them more efficient and scalable compared with traditional machine learning approaches. Hybrid architectures combining CNN and LSTM networks have also been proposed to capture both spatial and temporal characteristics of ECG signals, further improving diagnostic performance [9].

In addition to ECG analysis, echocardiography has become an essential imaging modality for assessing cardiac structure and function. Echocardiographic imaging enables clinicians to observe heart chamber size, ventricular function, valve movement, and wall motion abnormalities. Recent studies have explored the use of artificial intelligence to automatically interpret echocardiographic images.

Deep learning models trained on echocardiography datasets have demonstrated the ability to classify cardiac views and identify structural abnormalities with high accuracy. In particular, 3D-CNNs have shown strong performance in analyzing echocardiographic video sequences because they can capture spatio-temporal cardiac motion patterns. Studies using large echocardiography datasets have reported classification accuracies exceeding 96%, demonstrating the effectiveness of deep learning approaches in automated cardiac ultrasound interpretation [6]. These approaches reduce the burden on clinicians by automating image analysis and providing decision support during cardiovascular assessments.

Another important area of research focuses on integrating artificial intelligence with clinical decision support systems to improve early detection of heart disease. Large-scale studies have shown that AI-enhanced ECG analysis can identify structural heart disease and recommend further diagnostic tests, such as echocardiography, with higher accuracy than traditional clinical interpretation. Such systems have the potential to detect hidden cardiac conditions earlier and improve patient outcomes by guiding physicians toward appropriate diagnostic pathways [10].

Despite these advances, several limitations remain in existing research and healthcare systems. Many AI-based ECG studies focus primarily on arrhythmia classification rather than detecting conditions that mimic gastric symptoms or differentiating cardiac and non-cardiac causes of chest or epigastric pain. Similarly, many echocardiography-based studies concentrate on image classification rather than integrating ECG and echocardiographic findings within a comprehensive diagnostic workflow. Furthermore, existing systems often operate as isolated diagnostic tools rather than components of a broader clinical decision framework that incorporates multiple stages of validation.

Additionally, most current AI models rely on single-modality analysis, meaning they analyze either ECG signals or echocardiographic images independently. However, in real clinical practice, physicians often rely on multiple diagnostic tests to confirm cardiac abnormalities. The absence of integrated diagnostic systems that combine ECG and echocardiography analysis within a structured clinical workflow represents a significant gap in current research and healthcare technology. Addressing this limitation could significantly improve the reliability and accuracy of cardiac diagnosis in patients presenting with ambiguous symptoms.

1.3 Research Gap

Despite significant progress in artificial intelligence for cardiovascular diagnostics, several important limitations remain in the current research landscape, particularly in the combined analysis of electrocardiogram (ECG) waveform images and echocardiography images for early cardiac risk identification. Existing studies primarily focus on analyzing single diagnostic modalities, such as ECG signals or echocardiographic images independently, rather than integrating multiple cardiac diagnostic sources into a unified decision-support framework.

Recent studies have demonstrated the effectiveness of deep learning algorithms for automated ECG interpretation, particularly in arrhythmia detection and waveform classification. For example, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have been widely applied to detect abnormalities in ECG signals with high classification accuracy [11]. However, most of these approaches are designed for isolated ECG-based diagnosis, which limits their ability to capture structural cardiac abnormalities that are only visible through imaging modalities such as echocardiography.

Similarly, research on echocardiogram analysis has focused primarily on view classification and cardiac chamber segmentation, rather than integrating echocardiographic findings with physiological signal data [12]. Although these approaches improve the automation of echocardiography interpretation, they still operate as stand-alone diagnostic systems and do not consider complementary information from other cardiac diagnostic tools.

Another limitation observed in current literature is the lack of decision-support systems designed for educational or diagnostic workflow assistance. Most existing AI models are optimized for hospital diagnostic environments and are not structured to support clinical reasoning processes or assist medical students in understanding diagnostic pathways [13].

Furthermore, current research typically concentrates on detecting specific cardiac diseases, such as arrhythmia or heart failure, without addressing broader clinical scenarios where symptoms such as chest pain may originate from multiple causes, including gastrointestinal conditions. As a result, many AI diagnostic systems fail to support differential diagnosis between cardiac and non-cardiac conditions, which is a critical challenge in early clinical assessment [14].

Another significant research gap lies in the integration of ECG waveform images analysis and echocardiographic imaging within a single AI-based architecture. While both modalities individually provide valuable insights into cardiac health, limited research has explored how combining these two diagnostic sources can improve early cardiac abnormality detection and clinical decision support [15].

Therefore, there is a clear need for a multi-modal AI-assisted diagnostic component capable of simultaneously analyzing ECG waveform images and echocardiogram images. Such an approach would allow the system to capture both electrophysiological abnormalities and structural heart conditions, improving diagnostic accuracy and supporting more reliable clinical decision-making.

Furthermore, existing research rarely investigates the combined use of 2D-CNNs for ECG waveform image analysis and 3D-CNNs for echocardiographic video analysis within a unified diagnostic workflow. Integrating these complementary deep learning architectures could significantly improve early cardiac abnormality detection by capturing both electrical and structural characteristics of the heart.

The proposed Heart Signal & Functional Analysis Component (Component 2) addresses these limitations by integrating deep learning models for ECG signal processing and echocardiographic image analysis within a unified framework. By combining these two diagnostic modalities, the system aims to provide a more comprehensive assessment of cardiac function while supporting medical students in distinguishing cardiac-related chest pain from other potential causes.

Ref No	Research Study	Method / Approach	Key Contribution	Identified Limitation
[11]	Deep learning for automated ECG interpretation	CNN / RNN models for ECG classification	High accuracy in arrhythmia detection	Focuses only on ECG signals without integrating imaging data
[12]	AI-based echocardiographic view classification	CNN-based image classification	Improves automated echocardiogram interpretation	Limited integration with physiological signals
[13]	AI systems for cardiovascular diagnosis	Machine learning decision-support models	Assists physicians in detecting cardiac diseases	Not designed for educational or diagnostic workflow support
[14]	Chest pain diagnosis using AI methods	Clinical decision algorithms	Improves detection of cardiac-related chest pain	Does not address differential diagnosis with gastric conditions
[15]	Multi-modal cardiovascular diagnostic research	Integration of signal and imaging data	Demonstrates improved diagnostic accuracy when combining ECG and imaging modalities	Limited implementation within real-time clinical decision-support systems

Table 2: Summary of Research Evidence

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop an AI-assisted diagnostic component capable of analyzing ECG waveform images and echocardiographic data to detect cardiac abnormalities and support early differentiation between cardiac and gastric-related symptoms within a multi-stage clinical diagnostic workflow.

2.2 Specific Objectives

- To develop a deep learning - based ECG analysis model capable of detecting abnormal cardiac patterns from ECG images for early identification of potential cardiac conditions.
- To collect and preprocess publicly available ECG waveform image datasets and echocardiography datasets to prepare them for deep learning model training.
- To design and implement an echocardiography image analysis model using deep learning techniques to detect structural and functional abnormalities of the heart, such as wall motion abnormalities and ventricular dysfunction.
- To integrate ECG and echocardiographic analysis into a sequential diagnostic validation stage within the proposed clinical workflow to improve the reliability of cardiac diagnosis.
- To evaluate the performance of the developed AI models using appropriate metrics such as accuracy, precision, recall, and F1-score to ensure reliable diagnostic performance.
- To develop a system that provides interpretable outputs from ECG and Echo analysis that can support clinicians in making more accurate diagnostic decisions.

3. METHODOLOGY

3.1 Research Approach

This research follows an experimental research approach combined with a design science methodology. The experimental approach is used to train and evaluate deep learning models for analyzing ECG waveform report images and echocardiographic images, while the design science methodology focuses on designing and developing an intelligent diagnostic component that can solve a real-world healthcare problem. The study involves building machine learning models, training them using medical datasets, evaluating their performance, and integrating them into the proposed diagnostic workflow. This approach enables systematic experimentation with different algorithms and model architectures to identify the most effective solution for detecting cardiac abnormalities.

3.2 Technical Methods and Algorithms

The proposed component utilizes deep learning techniques for biomedical signal and image analysis. Two separate AI models will be developed for ECG and echocardiography analysis.

For ECG analysis, a 2D-CNN will be trained using ECG waveform image datasets. The 2D-CNN architecture is suitable for analyzing spatial patterns in ECG report images such as waveform morphology, ST-segment deviations, and rhythm abnormalities. The model will automatically learn important ECG features such as waveform shapes, ST-segment variations, and abnormal rhythm patterns that are associated with cardiac diseases. The training process will involve preprocessing ECG images, feature extraction using convolutional layers, and classification using fully connected layers.

For echocardiography analysis, a 3D-CNN will be implemented to analyze sequential ultrasound image frames of the heart. The 3D-CNN architecture enables the model to capture spatio-temporal cardiac motion patterns, which helps identify abnormalities such as ventricular dysfunction and wall motion abnormalities.

Both models will undergo training, validation, and testing phases using labeled datasets. The models will be trained using the Adam optimizer with a cross-entropy loss function for classification tasks. Data augmentation techniques such as rotation, brightness adjustment, and noise addition will be applied to improve model generalization. Model performance will be evaluated using standard medical AI metrics, including accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC). During training, optimization techniques such as backpropagation and stochastic gradient descent will be used to adjust model parameters and minimize prediction error.

3.3 Tools and Platforms

The proposed ECG and echocardiography analysis component will be developed using a set of modern artificial intelligence tools and software platforms designed for medical image analysis and deep learning model development.

Python will be used as the primary programming language for implementing the system due to its extensive support for machine learning, deep learning, and medical data processing libraries. Python provides a flexible and powerful development environment for implementing convolutional neural networks, performing data preprocessing, and handling large volumes of medical datasets.

The deep learning models for ECG image analysis and echocardiographic image classification will be developed using the PyTorch framework, which is widely used in research for building and training deep neural network architectures. PyTorch provides dynamic computational graphs, efficient GPU acceleration, and flexibility for implementing advanced deep learning techniques required for medical image analysis.

For image preprocessing and feature preparation, the OpenCV library will be used to perform operations such as image resizing, noise reduction, normalization, and waveform enhancement.

The Albumentations library will be used for advanced data augmentation techniques that improve the robustness of deep learning models trained on medical image datasets.

In addition, the NumPy library will be used for efficient numerical computations and manipulation of multidimensional image data during preprocessing and model training stages. The Pandas library will be used for dataset organization, annotation management, and preprocessing of metadata associated with ECG and echocardiography datasets.

Model experimentation and development will be conducted using Jupyter Notebook and Google Colab environments, which provide interactive programming interfaces for developing and testing machine learning models. These platforms also provide GPU acceleration, enabling faster training of deep learning models using large ECG and echocardiographic image datasets.

For visualization and performance evaluation, Matplotlib will be used to generate graphs such as training accuracy curves, loss curves, and confusion matrices. Additionally, the Scikit-learn library will be used to calculate evaluation metrics including accuracy, precision, recall, and F1-score to assess the performance of the trained models.

Together, these tools and platforms provide a comprehensive development environment that supports efficient implementation, training, evaluation, and visualization of the proposed AI-based cardiac diagnostic models.

3.4 Data Collection and Ethical Considerations

The datasets used for training the ECG and echocardiography models will consist of publicly available medical datasets and anonymized medical images obtained from approved medical sources. ECG datasets may include commonly used benchmark datasets such as the MIT-BIH Arrhythmia Dataset and PTB-XL Dataset, which contain ECG recordings that can be converted into ECG waveform report images for training image-based classification models. Echocardiographic datasets will include labeled ultrasound images or video sequences representing both normal and abnormal cardiac conditions, such as the EchoNet-Dynamic dataset used for cardiac function analysis.

All patient data used in this research will be fully anonymized to ensure privacy and confidentiality. No personally identifiable information will be collected or stored during the research process. Ethical guidelines for medical data usage will be strictly followed to ensure responsible handling of healthcare data. The research will comply with relevant data protection regulations and ethical standards in medical research.

The study will also follow institutional research ethics guidelines, and only publicly available or properly authorized datasets will be used for model training and evaluation.

3.5 Validation Strategy and Performance Metrics

The developed AI models will be evaluated using standard machine learning validation techniques. The dataset will be divided into training, validation, and testing subsets to ensure unbiased performance evaluation. Cross-validation techniques may also be used to improve reliability of the results.

Several performance metrics will be used to measure the effectiveness of the proposed models. These include accuracy, precision, recall, sensitivity, specificity, and F1-score, which evaluate how well the model correctly identifies cardiac abnormalities. In addition, confusion matrices will be used to visualize classification results and analyze the distribution of correct and incorrect predictions.

Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) values may also be used to measure the diagnostic capability of the models. These evaluation methods will help determine whether the developed models provide reliable and clinically useful diagnostic support.

The final performance results will be compared with existing research studies to determine whether the proposed approach provides improved diagnostic accuracy and reliability for cardiac abnormality detection using ECG and echocardiographic data.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Its Components

The proposed ECG and Echocardiography Analysis Component is an artificial intelligence–assisted diagnostic module designed to support the identification of potential cardiac abnormalities in patients presenting with chest pain symptoms. This component plays a critical role in distinguishing cardiac-related conditions from non-cardiac causes by analyzing cardiac diagnostic data using advanced machine learning techniques.

In the overall system architecture, this component operates after the Baseline Risk & Patient Profile Analysis module (Component 1) has evaluated patient questionnaire responses and demographic information. If the preliminary assessment indicates a possible cardiac risk, the system proceeds to Component 2 – Heart Signal & Functional Analysis, where cardiac diagnostic tests are analyzed.

This component processes two primary sources of cardiac diagnostic information: electrocardiogram (ECG) waveform images and echocardiography images. ECG waveform images represent the electrical activity of the heart and are widely used for detecting abnormal heart rhythms, ischemia, and other cardiac conditions. Echocardiography images provide structural and functional information about the heart, including chamber size, valve function, and blood flow patterns. Recent studies have demonstrated that deep learning models can effectively analyze echocardiography videos for cardiac function assessment [16].

Deep learning models are applied to analyze both ECG and echocardiographic data. A 2D-CNN is used to analyze ECG waveform images, while a 3D-CNN is applied to analyze echocardiographic ultrasound image sequences. The ECG analysis model focuses on identifying abnormal waveform patterns associated with potential cardiac conditions, while the echocardiography model examines structural abnormalities in cardiac imaging data. By combining these two diagnostic sources, the system improves the reliability of cardiac abnormality detection.

The outputs generated by this component include cardiac abnormality detection results, diagnostic indicators, and risk scores, which are then passed to the next stage of the diagnostic workflow. If significant cardiac abnormalities are detected, the system may trigger further investigation through Component 3 – Coronary Blockage Detection. If no cardiac abnormalities are identified, the workflow proceeds to Component 4 – Gastric Analysis & Decision Support Engine to investigate potential gastrointestinal causes of chest pain.

Through this integrated diagnostic process, the ECG and echocardiography analysis component enhances the system’s ability to simulate real-world clinical decision-making while supporting medical students in understanding cardiac diagnostic procedures.

4.2 Overall System Diagram

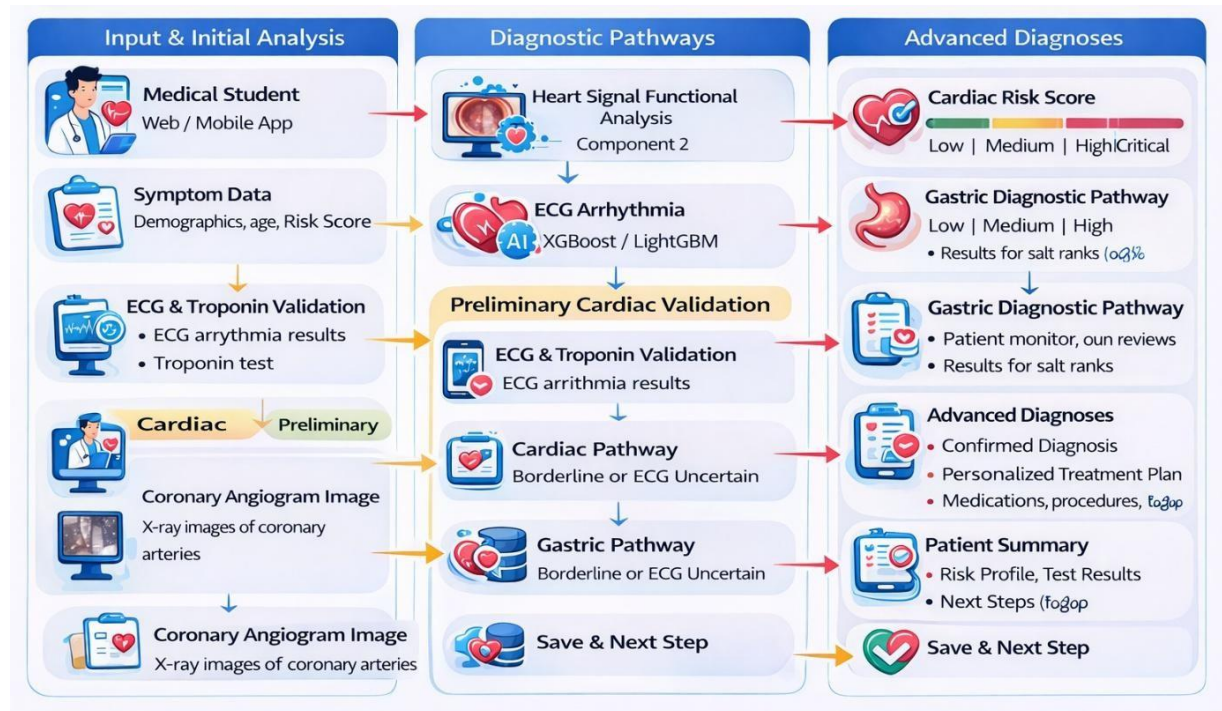


Figure 1: High-Level System Architecture Diagram

4.3 High-Level Architecture of Heart Signal & Functional Analysis

The ECG and Echocardiography Analysis Component consists of several internal modules responsible for handling different stages of cardiac diagnostic analysis.

The first module is the ECG Data Processing Module, which receives ECG waveform report images uploaded to the system. This module performs preprocessing tasks such as noise reduction, normalization, and signal transformation before feeding the data into the deep learning model for classification.

The second module is the ECG Pattern Classification Model, which uses convolutional neural networks to identify abnormal cardiac patterns within ECG waveforms. This model detects

indicators such as irregular rhythms, abnormal wave morphology, or potential signs of cardiac ischemia.

The third module is the Echocardiography Image Analysis Module, which processes ultrasound images of the heart. This module applies deep learning techniques, specifically 3D-CNNs, to analyze echocardiography image sequences and detect abnormalities related to heart chambers, valves, and ventricular function [17].

The final module is the Diagnostic Output Generator, which combines the results of the ECG and echocardiography analyses to produce a cardiac diagnostic assessment. The generated outputs include diagnostic indicators and risk scores that support further clinical evaluation.

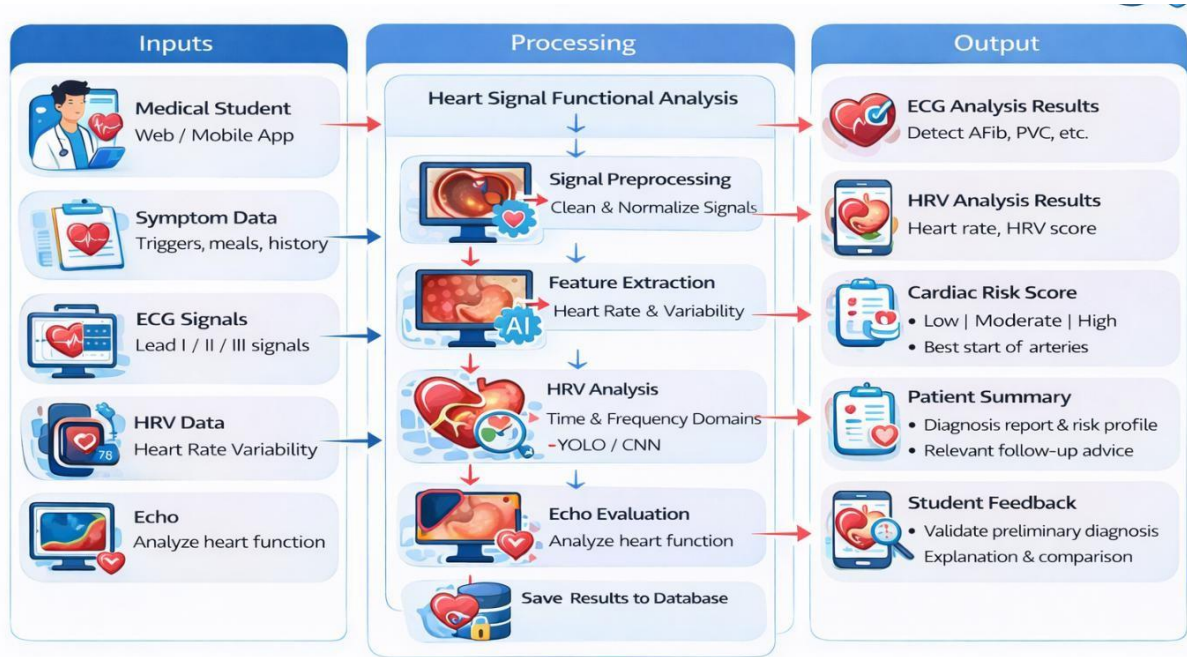


Figure 2: ECG and Echocardiography Analysis Component

4.4 Component Interactions

The ECG and echocardiography analysis component interacts closely with other modules within the overall diagnostic system. Initially, the system receives patient profile data from Component 1 - Baseline Risk & Patient Profile Analysis, which determines whether cardiac diagnostic testing is required.

Once activated, the ECG and echocardiography component processes cardiac diagnostic data and generates analysis results. These results are then forwarded to Component 3 - Coronary Blockage Detection, where additional analysis of coronary angiography images may be performed if cardiac abnormalities are suspected.

If the analysis results indicate that cardiac conditions are unlikely, the system passes the workflow to Component 4 - Gastric Analysis & Decision Support Engine, where gastric

symptoms and gastroscopy images are analyzed to determine potential gastrointestinal causes of chest pain.

This structured interaction between components ensures that the diagnostic workflow follows a logical medical decision-making process.

4.5 Data Flow Explanation

The data flow within the ECG and echocardiography analysis component begins when the system receives cardiac diagnostic inputs such as ECG waveform images and echocardiography scans. These inputs are first processed through data preprocessing steps, including noise removal, image resizing, normalization, and augmentation.

Once preprocessing is complete, the data is fed into trained deep learning models designed to detect abnormal cardiac patterns and structural heart abnormalities. ECG waveform images are analyzed using convolutional neural networks, while echocardiography sequences are processed using 3D convolutional architectures capable of learning spatial and temporal cardiac features [17].

After model inference is completed, the system generates diagnostic outputs such as cardiac abnormality indicators, confidence scores, and classification results. These outputs are then transmitted to the system's central decision logic, which determines whether additional cardiac analysis or gastric analysis should be performed.

This structured data flow ensures efficient processing of medical data while supporting accurate diagnostic evaluation.

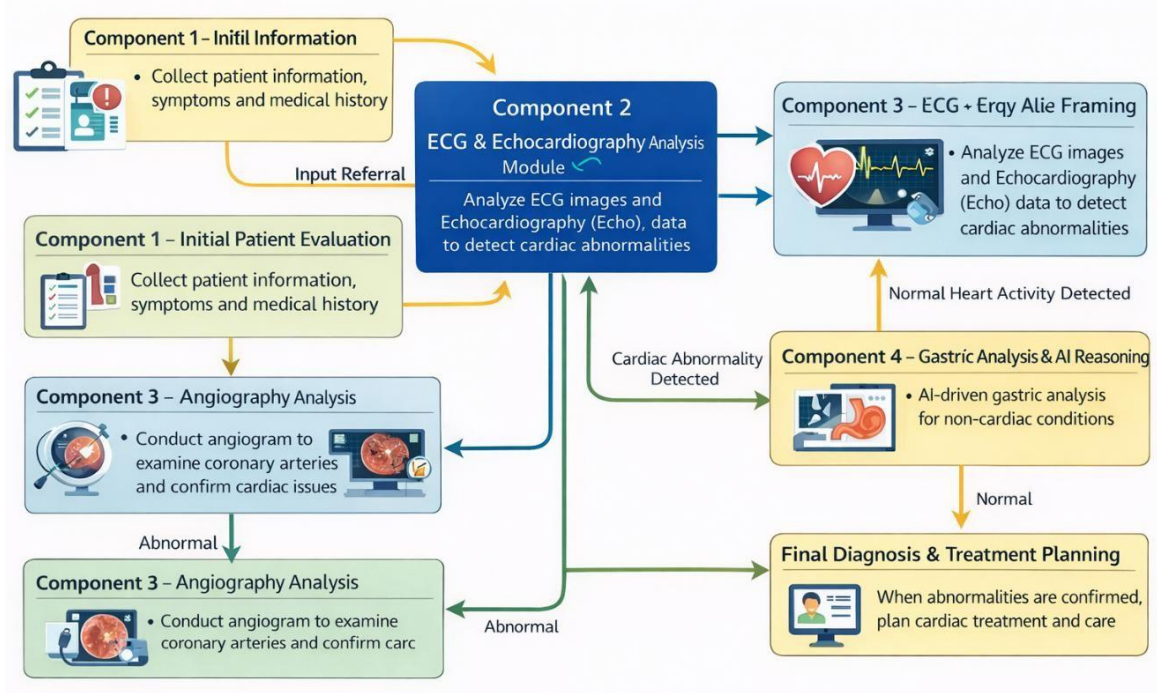


Figure 3: ECG and Echocardiography Analysis Data Flow Diagram

4.6 Scalability and security considerations

The proposed architecture is designed to support scalability and secure handling of medical data. The system can be deployed using cloud-based infrastructure, allowing computational resources to scale according to the volume of uploaded ECG and echocardiography data. Cloud deployment also enables remote accessibility, which is beneficial for educational environments and research demonstrations.

To protect sensitive medical information, the system incorporates several security mechanisms, including secure data transmission, encrypted storage, and controlled access to system components. Authentication and authorization mechanisms ensure that only authorized users such as medical students, instructors, or administrators can access the diagnostic platform.

Additionally, anonymization techniques can be applied to patient data before processing to ensure compliance with ethical standards and data privacy requirements in medical research environments.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed AI-assisted cardiac analysis component involves several stakeholders who interact with the system during the diagnostic learning and analysis process. The primary stakeholders are medical students, who will use the system as an educational decision-support tool to upload and analyze ECG images and echocardiographic scans to understand how cardiac abnormalities are identified through diagnostic data. This component enables students to explore the interpretation of ECG waveforms and echocardiographic structures while learning how cardiac-related chest pain can be distinguished from other conditions.

Secondary stakeholders include medical instructors and clinical supervisors, who may utilize the system as a teaching support platform in medical education. Through the AI-generated analysis results of ECG and echocardiography data, instructors can demonstrate diagnostic reasoning, explain cardiac abnormalities, and evaluate the students' understanding of cardiac diagnostic processes.

In addition, system administrators and researchers are responsible for maintaining the AI models, updating ECG and echocardiography datasets, and monitoring the performance of the trained diagnostic models. These stakeholders ensure that the system remains accurate, reliable, and up-to-date with current medical knowledge.

From the clinical perspective, cardiologists and healthcare professionals may also benefit from the system by exploring AI-assisted interpretations of ECG waveform images and echocardiographic images for educational and research purposes. The outputs generated by the ECG and echocardiography analysis component provide cardiac abnormality detection results and diagnostic insights that support both medical learning and clinical research in cardiovascular disease identification.

5.2 Functional Requirements

1. ECG Data Input and Processing

- Allow users to upload or input ECG images obtained from ECG diagnostic tests.
- Automatically preprocess ECG images to prepare them for AI analysis.

2. ECG Abnormality Detection

- Analyze ECG images using a trained deep learning model.
- Detect abnormal cardiac patterns such as ST-segment abnormalities or irregular waveforms.

3. Echocardiography Image Analysis

- Allow users to upload echocardiographic ultrasound images of the heart.
- Analyze these images using an AI model to detect structural and functional cardiac abnormalities.

4. Sequential Diagnostic Validation

- Use ECG analysis as the first validation step for detecting potential cardiac conditions.
- Trigger echocardiographic analysis when ECG results indicate possible cardiac abnormalities.

5. Diagnostic Result Generation

- Generate a classification result indicating whether cardiac abnormalities are detected.
- Provide supporting diagnostic outputs such as probability scores and predicted conditions.

6. Clinical Decision Support Output

- Display AI-generated diagnostic insights to assist clinicians in interpreting results.
- Provide clear visual outputs that help medical students make faster and more accurate decisions.

7. Data Storage and Retrieval

- Store analyzed ECG and echocardiography data securely for future reference.
- Allow authorized users to retrieve previous diagnostic results when necessary.

5.3 Non-functional Requirements

Non-functional requirements describe the quality attributes and operational constraints of the proposed system.

1. Security Requirements

The system must ensure the confidentiality and privacy of patient medical data. Access to the system should be restricted through secure authentication mechanisms to ensure that only authorized healthcare professionals can view or analyze patient data. Additionally, data transmission and storage should follow healthcare data protection standards to prevent unauthorized access or data breaches.

2. Performance Requirements

The system should provide fast and efficient analysis of ECG and echocardiographic images to support real-time or near real-time clinical decision-making. AI model inference should produce results within a short time frame to ensure that the system can be used effectively in clinical environments where timely diagnosis is critical.

3. Usability Requirements

The system should provide a user-friendly interface that allows healthcare professionals to easily upload diagnostic data, view analysis results, and interpret outputs. The interface should be simple and intuitive to ensure that clinicians can use the system without requiring extensive technical training.

4. Reliability Requirements

The system must ensure consistent and reliable performance when analyzing medical data. AI models should produce stable results across different datasets and maintain high diagnostic accuracy during operation.

5. Scalability Requirements

The system should be capable of handling increasing amounts of medical data and multiple user requests without performance degradation. This ensures that the system can be expanded and deployed across larger healthcare institutions if required.

6. Maintainability Requirements

The system architecture should allow easy updates and improvements to AI models, datasets, and system components. This ensures that the system can adapt to new research developments and medical diagnostic requirements.

7. Compatibility Requirements

The system should be compatible with commonly used medical data formats and hospital information systems, enabling integration with existing healthcare technologies and clinical workflows.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed AI-assisted diagnostic system is primarily targeted at healthcare institutions, medical education institutes, and digital health technology providers. Hospitals and clinics that frequently manage patients with chest pain can benefit from the system by improving early diagnostic support and reducing unnecessary cardiac investigations. Medical universities and training hospitals represent another key market segment because the system can function as an educational decision-support tool for medical students.

The primary customer persona includes medical students, clinical trainees, and healthcare professionals who require decision-support tools during diagnostic evaluation. Secondary users include hospital administrators and healthcare technology providers who seek AI-based solutions to improve diagnostic efficiency and reduce operational costs.

6.2 Value Proposition

The proposed system delivers an AI-powered educational and diagnostic support platform that enables users to analyze ECG signals and echocardiographic images using advanced machine learning techniques. The integration of automated ECG pattern detection and echocardiographic analysis provides a comprehensive understanding of cardiac abnormalities.

The key value proposition of the system lies in its ability to bridge the gap between theoretical medical knowledge and practical diagnostic interpretation. By allowing students to upload real ECG and echo images and receive AI-generated analysis results, the system helps learners better understand how cardiovascular diseases are identified in clinical practice.

This combination of diagnostic support and learning assistance provides significant value for both clinical practice and medical education.

6.3 Revenue Model

The commercialization strategy for the proposed system follows a Software-as-a-Service (SaaS) model, where educational institutions and healthcare organizations can subscribe to the system through cloud-based access. Universities and teaching hospitals may purchase institutional subscriptions that allow multiple students and instructors to access the platform.

Another possible revenue model is a license-based model, where hospitals or medical training centers can purchase a licensed version of the system for internal use. Additionally, per-user subscription plans can be offered to medical students or healthcare professionals who wish to access the system individually for learning and training purposes.

6.4 Cost Estimation

The development and deployment of the proposed system involve several cost components including computing infrastructure, software tools, dataset preparation, and development resources. Hardware resources are required for training deep learning models using ECG and echocardiography image datasets. Cloud computing services are necessary to provide scalable computing power for model training and deployment.

Software costs include development tools, machine learning frameworks, and image processing platforms used for model development and system implementation. Human resource costs involve the efforts of machine learning developers, software engineers, and medical domain experts who contribute to the development, training, and validation of the AI models.

6.5 Pricing Strategy

The pricing strategy for the proposed system can be based on a subscription-based model. Healthcare institutions may be offered tiered pricing plans depending on the number of users and the level of system access required. For example, small clinics may subscribe to a basic diagnostic support package, while larger hospitals may use a full-featured enterprise version that includes advanced analytics and integration with hospital information systems.

For medical universities, an educational licensing model may be applied where institutions pay a yearly fee to provide system access for students and academic staff.

6.6 Competitive Advantage

The proposed system offers several advantages compared to traditional medical learning tools and diagnostic systems. Unlike conventional educational resources that rely on static textbooks or manual ECG interpretation training, this system provides AI-driven interactive analysis of ECG and echocardiographic images.

Another key advantage is the integration of multiple cardiac diagnostic modalities, where ECG waveform analysis is combined with echocardiographic image interpretation. This multi-level analysis approach allows users to gain a deeper understanding of cardiovascular diagnostics.

Additionally, the system supports automated analysis, faster diagnostic interpretation, and interactive learning, making it a valuable tool for both medical education and healthcare research.

6.7 Intellectual Property Considerations

The intellectual property of the proposed system includes the AI algorithms, software architecture, and integrated diagnostic workflow developed during the research. These components may be protected through software copyrights and potential patent applications related to the AI-based diagnostic decision framework.

Data privacy and ethical considerations must also be addressed when handling medical data. All patient information used for model training and system testing must be anonymized and handled according to healthcare data protection regulations. Licensing agreements may also be used to protect proprietary algorithms and prevent unauthorized system use.

7. BUDGET AND JUSTIFICATION

7.1 Estimated Budget Breakdown

Cost Category	Description	Estimated Cost (LKR)
Hardware	Use of personal laptop/workstation for ECG and echocardiography AI model development and testing	40,000
Software Tools	Development tools, IDE licenses, and development environment setup	5,000
Cloud Services	Cloud server for storing ECG datasets and deploying the AI models	30,000
Data Processing	Preparation, annotation, and preprocessing of ECG images and echocardiography datasets	10,000
Human Effort	Development, training, testing, and integration of ECG and echocardiography analysis models	15,000

Table 3: Overall Budget Breakdown Table

Total Estimated Budget: **100,000 LKR**

7.2 Hardware Cost Justification

The development of the ECG and Echocardiography Analysis Component requires a computer system capable of running deep learning frameworks and processing medical image datasets. A personal laptop or workstation will be used for model training, image preprocessing, and system development tasks.

Hardware resources are required for:

- Running deep learning frameworks such as PyTorch
- Training convolutional neural network models for ECG image classification
- Processing echocardiography images to identify cardiac abnormalities
- Running development environments, backend services, and testing tools

Since the research will utilize existing personal hardware for model development and experimentation, only a portion of the budget is allocated to represent hardware usage, maintenance, and computing resource consumption during the project development period.

7.3 Software Requirements

Software Tool	Purpose	Estimated Cost (LKR)
Python Programming Language	Development of AI models and backend services	Free
PyTorch	Implementation of deep learning models for ECG and echocardiography analysis	Free
OpenCV / Albumentations	Image preprocessing, augmentation, and data transformation	Free
FastAPI	Backend API development for system integration	Free
PostgreSQL	Database management for storing patient data and model outputs	Free
Development Tools (IDE, Git)	Coding, debugging, and version control	5,000

Table 4: Software Requirements for System Development

Justification:

Most software technologies used in this research are open-source frameworks, which significantly reduce development costs. These tools are widely used in machine learning and medical image analysis and provide sufficient capabilities to implement the proposed system.

7.4 Cloud Services

Service	Purpose	Estimated Cost (LKR)
Cloud Server Hosting	Hosting backend APIs and AI models	20,000
Cloud Storage	Storage of ECG datasets, echocardiography images, and system outputs	7,000
Deployment Services	Deployment, monitoring, and remote system access	3,000

Table 5: Cloud Services Cost

Justification:

Cloud services provide scalable computing infrastructure required for training and deploying machine learning models. The use of cloud platforms allows the AI models to be hosted remotely so that the system can be accessed during testing, evaluation, and demonstration phases.

Cloud storage is also essential for managing large ECG image datasets and echocardiography scans, ensuring secure and reliable data storage throughout the research project. Deployment services enable continuous monitoring and accessibility of the developed diagnostic system.

7.5 Human Effort Cost

Human effort cost represents the technical expertise and development time required to design, implement, and validate the ECG and echocardiography analysis models within the system.

This includes:

- Developing deep learning models for ECG waveform image classification
- Implementing echocardiography image analysis algorithms to detect cardiac abnormalities
- Performing image preprocessing and augmentation techniques to improve model performance
- Integrating the ECG and echo analysis component with the system backend
- Conducting model training, validation, and performance evaluation
- Testing the component within the overall diagnostic workflow

The estimated cost reflects the research effort, technical development, and experimentation required to successfully complete this component within the project timeline.

Activity	Description	Estimated Cost (LKR)
System Design	Designing questionnaire structure, risk scoring rules, and system architecture	4,000
AI Model Development	Developing and training machine learning models for cardiac and gastric risk prediction	5,000
Backend Development	Implementation of FastAPI backend services and database integration	3,000
Frontend Development	Development of the web-based dashboard interface using React	2,000
System Testing & Validation	Testing the system functionality and evaluating model performance	2,000
Research Travel	Transportation expenses for meetings, consultations, or field observations related to the study	2,000
Documentation & Reporting	Preparation of research reports, system documentation, and proposal writing	2,000

Table 6: Human Effort and Research Activity Cost Estimation

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Work Breakdown Structure Diagram

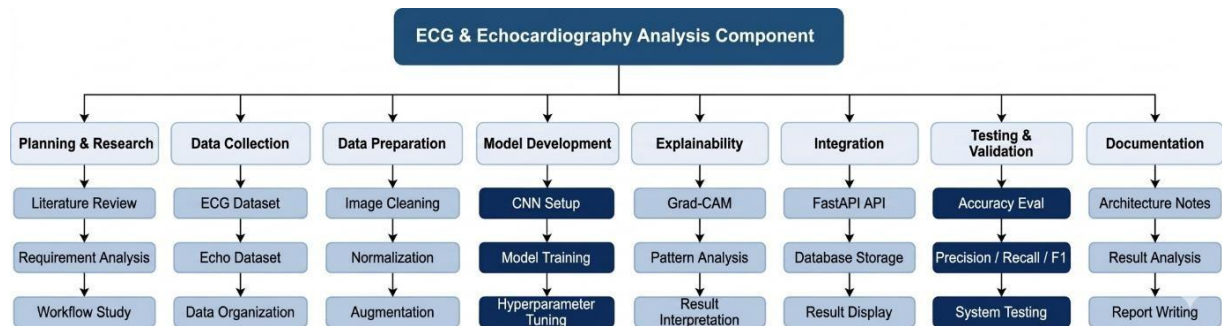


Figure 4: Work Breakdown Chart

This diagram illustrates the hierarchical decomposition of the research project into smaller, manageable tasks. The project activities are divided among team members based on their respective system components and responsibilities.

8.2 Task Breakdown Per Member

Task ID	Task Name	Description	Timeline
T1	Literature Review	Study research papers related to ECG signal analysis, echocardiography imaging, and deep learning techniques used for cardiovascular disease detection	March 2026
T2	Dataset Collection	Collect ECG image datasets and echocardiography scan datasets from public medical repositories and research datasets.	March - April 2026
T3	Dataset Preparation & Annotation	Organize and label ECG and echocardiography images according to diagnostic categories such as normal and abnormal cardiac patterns.	April 2026
T4	Image Preprocessing	Perform preprocessing tasks including image resizing, normalization, noise removal, and augmentation using image processing libraries.	April - May 2026
T5	ECG Analysis Model Development	Develop a deep learning model to classify ECG waveform images and identify abnormal cardiac patterns.	May - June 2026
T6	Echocardiography Analysis Model	Implement an AI model to analyze echocardiography images and detect structural or functional cardiac abnormalities.	June 2026

T7	Model Optimization	Perform hyperparameter tuning and performance improvement techniques to increase model accuracy and reliability.	June - July 2026
T8	Backend API Development	Develop backend APIs to process ECG and echocardiography inputs and communicate with trained AI models.	July - August 2026
T9	System Integration	Integrate the ECG and echocardiography analysis module with the overall diagnostic system and database.	August 2026
T10	System Testing and Validation	Evaluate model performance using metrics such as accuracy, precision, recall, and F1-score and validate the diagnostic workflow.	September - October 2026

Table 7: Work Breakdown Structure (WBS)

8.3 Timeline Alignment

The development tasks for the ECG and Echocardiography Analysis Component are aligned with the overall research project timeline to ensure that model development, system integration, and validation are completed before the final submission deadlines.

Phase	Period	Activities
Phase 1	March - April 2026	Literature review, dataset collection, and dataset organization
Phase 2	April - May 2026	Image preprocessing and dataset preparation
Phase 3	May - June 2026	ECG model development and echocardiography analysis model implementation
Phase 4	June - July 2026	Model optimization and hyperparameter tuning
Phase 5	July - August 2026	Backend API development and system integration
Phase 6	August - October 2026	System testing, validation, and performance evaluation

Table 8: Project Development Timeline for ECG And Echocardiography Analysis

8.4 Balanced Workload Distribution

The workload for the ECG and Echocardiography Analysis Component is organized into multiple development stages to ensure a balanced distribution of effort throughout the research lifecycle.

The first stage focuses on research and dataset preparation, including literature review, dataset collection, and organization of ECG and echocardiography images. This stage ensures that sufficient high-quality medical data is available for training the deep learning models.

The second stage focuses on image preprocessing and model development, where deep learning algorithms are implemented to analyze ECG waveform images and echocardiographic scans. During this stage, the models are trained to identify cardiac abnormalities and distinguish between normal and abnormal patterns.

The third stage involves system integration, where backend APIs are developed to connect the trained AI models with the overall diagnostic system and database infrastructure. This stage ensures smooth communication between the different system components.

The final stage focuses on testing, evaluation, and optimization, where model performance is measured using evaluation metrics such as accuracy, precision, recall, and F1-score. System testing is conducted to verify that the entire workflow operates correctly before the final research submission.

This structured development process ensures that the research progresses systematically from data preparation to AI model development, system integration, and final evaluation, enabling the successful implementation of the ECG and echocardiography diagnostic component.

9. GANTT CHART

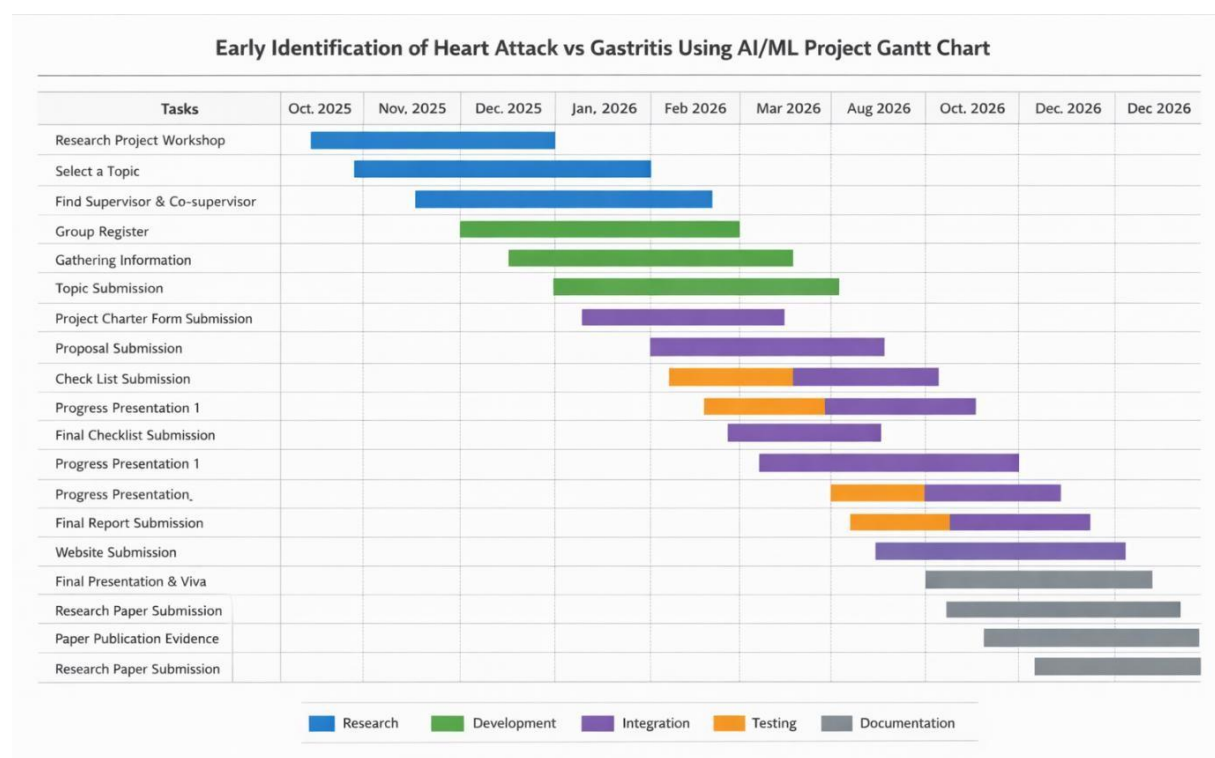


Figure 5: Grantt Chart

REFERENCES

- [1] S. Mahmood, R. A. Rashid, and M. Ali, “Atypical myocardial infarction masquerading as gastroenteritis,” *International Journal of Research in Medical Sciences*, 2023.
- [2] Health and Disability Commissioner, “Misdiagnosis of acute heart disease as gastro-oesophageal reflux,” 2005.
- [3] S. Bösner et al., “Heartburn or angina? Differentiating gastrointestinal disease in primary care patients presenting with chest pain,” *International Archives of Medicine*, 2009.
- [4] O. M. P. Jolobe, “Differential diagnosis of gastrointestinal symptoms and ST segment elevation,” 2021.
- [5] Z. Wu and C. Guo, “Deep learning and electrocardiography for cardiovascular disease diagnosis,” *BioMedical Engineering Online*, 2025.
- [6] J. A. Naser et al., “Artificial intelligence-based classification of echocardiographic views,” *European Heart Journal – Digital Health*, 2024.
- [7] M. Khalid et al., “Applications of deep learning in ECG classification for disease diagnosis,” *Engineering Journal*, 2024.
- [8] T. J. Jun et al., “ECG arrhythmia classification using a 2-D convolutional neural network,” 2018.
- [9] A. Peimankar and S. Puthusserypady, “DENS-ECG: Deep learning approach for ECG signal delineation,” 2020.
- [10] AI-assisted ECG analysis for structural heart disease detection, *Nature research findings*, 2025.
- [11] R. Hannun et al., “Cardiologist-level arrhythmia detection using deep neural networks,” *Nature Medicine*, 2019.
- [12] J. Zhang et al., “Fully automated echocardiogram interpretation using deep learning,” *Nature*, 2018.
- [13] S. Rajpurkar et al., “Deep learning for automated ECG diagnosis,” *npj Digital Medicine*, 2020.
- [14] P. Swapna et al., “Automated detection of cardiovascular disease using machine learning techniques,” *Healthcare Technology Letters*, 2019.
- [15] A. Madani et al., “Fast and accurate view classification of echocardiograms using deep learning,” *npj Digital Medicine*, 2018.
- [16] Ouyang, D. et al., “Video-based AI for beat-to-beat assessment of cardiac function,” *Nature*, 2020.

[17] Tran, D. et al., “Learning spatiotemporal features with 3D convolutional networks,” *Proceedings of the IEEE International Conference on Computer Vision*, 2015.

10. APPENDICES

Appendix A – Survey Instruments

This appendix contains the questionnaire used to collect patient symptom data related to chest pain and gastric conditions. The survey includes questions regarding patient demographics, lifestyle factors, meal-related symptoms, medication history, and other relevant clinical indicators that help differentiate cardiac and gastric chest pain.

Appendix B – Preliminary UI Mockups

This appendix includes preliminary user interface designs for the medical student dashboard. The mockups demonstrate how users can upload gastroscopy images, review AI predictions, view severity levels, and access treatment recommendations generated by the system.

Appendix C – Risk Analysis

Potential Risk	Description	Mitigation Strategy
Limited Dataset Availability	Insufficient endoscopy images for training the AI model may affect model performance.	Use publicly available datasets such as HyperKvasir and apply data augmentation techniques.
Model Accuracy Issues	The AI model may produce incorrect predictions due to limited or imbalanced data.	Use validation techniques, improve dataset quality, and tune model parameters.
System Integration Problems	Difficulty integrating the gastric analysis module with other system components.	Use modular architecture and well-defined APIs for communication between components.
Data Privacy Concerns	Handling medical images and patient data may raise privacy issues.	Ensure all datasets are anonymized and implement secure data storage methods.
Hardware or Resource Limitations	Limited computing power may slow down model training and testing.	Use cloud-based computing resources and optimize model training processes.

Table 9: Risk Analysis

Appendix D – Ethical Documents

This appendix includes ethical considerations related to the use of medical datasets and patient information. All patient data used in the research will be anonymized to protect privacy, and datasets will be used strictly for research and educational purposes in accordance with medical data protection guidelines.